Tutorial Questions | Week 11

COSC2779 - Deep Learning

This tutorial reviews model interpretation: post-hoc explanation methods (Grad-CAM, LIME, SHAP, Integrated Gradients), interpretable-by-design models (concept bottlenecks), and explaining Transformers and vision–language models. Please try the questions before you join the session. The question below has a similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a dermatology clinic network that uses a deep learning system to help triage skin lesions. The system has three parts:

   - a **CNN** that classifies dermoscopy images as *melanoma* or *benign*;
   - a **Transformer** text classifier that reads GP referral letters and flags *urgent* referrals;
   - a **vision–language model (VLM)** that is being trialled to draft a short report for each image.

   The clinic's governance committee will not approve deployment until the team can explain why the models make their predictions and show that those explanations can be trusted.

   (a) Why does interpretability matter particularly in this setting? Explain the difference between a **local** and a **global** explanation, giving one example of each for the melanoma classifier.

   > **Solution:**
   >
   > - High-stakes medical decisions: clinicians must be able to check the reasoning, detect errors and bias, and justify decisions; regulators and patients expect accountability.
   > - **Local**: explains a single prediction – e.g. a Grad-CAM heatmap showing which region of this image led to "melanoma".
   > - **Global**: describes the model's behaviour overall – e.g. TCAV showing the concept "irregular border" consistently increases the melanoma score, or feature visualisation of what filters respond to.

   (b) When the team applies **Grad-CAM** to melanoma predictions, the heatmaps often highlight a ruler and ink markings placed next to lesions, rather than the lesion itself. What does this reveal, and what would you do about it?

   > **Solution:** The model has learned a **shortcut / spurious correlation**: suspicious lesions are more often photographed with rulers or skin markings, so the model uses them as evidence. It would fail on images where this correlation does not hold.
   >
   > - Clean or crop the images, or mask out the artefacts; collect data where markings appear for both classes; augment with synthetic markings.
   > - Re-train and re-check with the explanation method; evaluate performance on images with and without markings.
   >
   > This is a key use of explanations: debugging the model, not just justifying it.

   (c) The classifier will be supplied by a vendor as an **API**: you can send images and receive probabilities, but have no access to weights or gradients. Can you still use Grad-CAM? Describe how **LIME** would explain a prediction in this setting.

   > **Solution:** No – Grad-CAM needs internal feature maps and gradients. A **model-agnostic** method is required. LIME:
   >
   > 1. Segment the image into **superpixels**.
   > 2. Create many perturbed images by randomly switching superpixels off (e.g. greyed out); encode each as a binary vector z.
   > 3. Send them to the API and record the melanoma probabilities.

4. Weight each sample by its closeness to the original image.
5. Fit a weighted sparse linear model $g(z)$; the superpixels with the largest positive weights are the explanation.

(d) To study an explanation more carefully, the team splits an image into two regions: the lesion **border** (A) and the lesion **colour pattern** (B). With the other region replaced by a neutral baseline, the model gives:

$v(\varnothing)$	$v(\{A\})$	$v(\{B\})$	$v(\{A, B\})$
0.1	0.5	0.3	0.9

Compute the **Shapley values** ϕ_A and ϕ_B, and the **leave-one-out** scores for each region. Compare the two and comment.

Solution: Shapley values average the marginal contribution over both orderings:

$$\phi_A = \tfrac{1}{2}\big[(v(\{A\}) - v(\varnothing)) + (v(\{A, B\}) - v(\{B\}))\big] = \tfrac{1}{2}[0.4 + 0.6] = \mathbf{0.5}$$
$$\phi_B = \tfrac{1}{2}\big[(v(\{B\}) - v(\varnothing)) + (v(\{A, B\}) - v(\{A\}))\big] = \tfrac{1}{2}[0.2 + 0.4] = \mathbf{0.3}$$

Check (efficiency): $0.5 + 0.3 = 0.8 = v(\{A, B\}) - v(\varnothing)$.

Leave-one-out: A: $0.9 - 0.3 = 0.6$; B: $0.9 - 0.5 = 0.4$; total $1.0 \neq 0.8$.

Both rank the border as more important, but leave-one-out **over-counts**: the two regions interact (together they add more than separately), and LOO gives each the full interaction effect. SHAP shares it fairly and its scores add up exactly to the change in prediction.

(e) In both LIME and SHAP, "removing" a region means replacing it with something (e.g. a black patch or a blur). Why can this choice make an explanation misleading, and how could you reduce the problem?

Solution: Images with black patches are **out-of-distribution**: the model never saw anything like them during training, so its response may reflect the artificial patch, not the missing information. The explanation then describes the model's reaction to the perturbation rather than its reasoning on real images.

- Use more natural baselines: blur, the mean colour, or in-painting; or replace regions with content from a background dataset.
- Compare several baselines – conclusions that change with the baseline should not be trusted.
- Where possible, train with masking augmentation so "removed" regions are in-distribution.

(f) The dermatologists would prefer a model that reasons like they do using the **7-point checklist** (pigment network, streaks, blue-whitish veil, ...), and that they can correct. Propose an **interpretable-by-design** model. Describe its structure, how a clinician could intervene, and one limitation.

Solution: A **Concept Bottleneck Model (CBM)**:

- Stage 1, g: image $\rightarrow$ predicted concept scores $\hat{c}$ (one per checklist feature).
- Stage 2, f: concepts $\rightarrow$ diagnosis, $\hat{y} = f(g(x))$. The prediction depends on the image only through the concepts, so the concept layer is the explanation.
- **Test-time intervention**: if the clinician sees a concept is wrong (e.g. "no streaks" when streaks are present), they overwrite that concept value and f recomputes the diagnosis.

Limitations: requires concept labels for every training image (expensive expert annotation); the concept set may be incomplete, costing accuracy; soft concept scores can leak extra information, so the concepts may not tell the whole story.

(g) For the referral-letter Transformer, a colleague shows a map of the model's **attention weights** as the explanation of why a letter was flagged urgent. Is this justified? Suggest better alternatives for explaining a Transformer.

Solution: Only partly. Raw attention shows **where information flows**, not **how much each token mattered** to the output: it ignores values, residual connections and feed-forward layers, a single head/layer is only part of the picture, and very different attention patterns can give the same prediction ("attention is not explanation" debate). Better:

- **Attention rollout**: combine attention across all layers (including residual connections).
- **Gradient-weighted attention** (Chefer et al.): weight attention by its gradient – class-specific, like Grad-CAM.
- **Integrated Gradients** or **SHAP/LIME** over words for token-level attribution.

(h) The team first tries **gradient** $\times$ **input** on the text model, but for letters the model is very confident about, all token scores are almost zero. Why? Explain how **Integrated Gradients** addresses this, including the role of the baseline.

Solution: When the model is very confident the output softmax is **saturated**: small changes to the input barely change the output, so the local gradient is close to zero even for important tokens.

Integrated Gradients accumulates gradients along a path from a **baseline** e' (e.g. all tokens replaced by `[PAD]`) to the input e:

$$R_i = (e_i - e'_i) \cdot \int_0^1 \frac{\partial f\big(e' + \alpha(e - e')\big)}{\partial e_i}\, d\alpha.$$

Along the path it passes through the non-saturated region where gradients are informative. Its scores satisfy **completeness**: they sum to $f(e) - f(e')$. The baseline defines "relevant compared to what", so different baselines can give different explanations.

(i) When the VLM is asked "*Is this lesion suspicious? Explain your answer*", it gives a fluent, clinically worded justification. The governance committee asks: "Doesn't this give us explanations for free?" Respond, and explain how the type of access to the VLM (open weights vs a closed API) affects what you can do.

Solution: The text explanation costs only a prompt, but it is **not guaranteed to be faithful**: it is generated text, optimised to sound right, and may describe evidence the model did not actually use (or hallucinate findings). Plausible $\neq$ faithful; it should be treated as a claim to test.

- **Open weights**: can compute attention/relevance maps from text tokens to image patches, gradients, Integrated Gradients, probing, activation patching – and use them to check the self-explanation.
- **Closed API**: limited to black-box perturbation (LIME/SHAP-style masking) and self-explanations.

(j) Several explanation methods now produce different heatmaps for the same images. Describe two tests the team could use to decide which explanations are **faithful** to the model.

Solution:

- **Deletion test**: remove the regions/tokens in order of attributed importance and track the prediction. A faithful explanation makes the score drop fastest; compare with a random ordering.
- **Model randomisation (sanity check)**: randomise the model's weights and recompute. If the explanation barely changes, it reflects the input (e.g. edges) rather than the model.
- Also: counterfactual checks (edit the cited evidence, e.g. remove the streaks, and see if the prediction changes), and agreement across methods.